



**LUND**  
UNIVERSITY

**Written Examination**  
**Linear Analysis, MATB24**  
**March 17, 2021**  
**Time: 09.00–16.00**

Centre for Mathematical Sciences  
Mathematics, Faculty of Science

*Hand in electronically in the form of a PDF-file no later than at 15.59. You may freely use formulas from the formula sheet on the canvas page. Refer to relevant theorems from the course when you use them. Write clearly with short and concise motivations.*

1. Find a power series solution to the problem

$$y''(x) - 2xy'(x) - 2y(x) = 0, \quad y(0) = 1, \quad y'(0) = 0.$$

Also write the solution in closed form.

2. (a) Determine the left-handed limit

$$\lim_{x \rightarrow 1^-} \sum_{n=0}^{\infty} n i^n x^n.$$

- (b) Compute the limit

$$\lim_{n \rightarrow \infty} \int_0^1 x^{10} \cos^4(nx) dx.$$

3. Determine a  $2\pi$ -periodic function  $y(x)$  such that

$$\pi y'(x) + y(x - \pi) = \cos^2 x.$$

Also determine the integral

$$\int_{-\pi}^{\pi} y(x)^2 dx.$$

4. Determine a function  $f(x)$  such that

$$\int_{-\infty}^{+\infty} (3 - 4y^2) e^{-y^2} f(x - y) dy = e^{-x^2}, \quad x \in \mathbb{R}.$$

5. (a) Let  $f$  be the  $2\pi$ -periodic function such that  $f(x) = |x|$ ,  $-\pi \leq x \leq \pi$ . Develop  $f$  in a Fourier series.  
(b) Find a continuous function  $u(x, y)$  on the closed disc  $x^2 + y^2 \leq 1$  which solves the Dirichlet problem  $u_{xx} + u_{yy} = 0$  in the unit disc  $x^2 + y^2 < 1$  with boundary values  $u(\cos \theta, \sin \theta) = |\theta|$  when  $-\pi \leq \theta \leq \pi$ . You may express your answer in terms of polar coordinates.  
(c) Find a power series  $f(z) = \sum_{n=0}^{\infty} a_n z^n$  with radius of convergence at least 1 such that  $u(x, y) = \operatorname{Re} f(z)$  when  $z = x + iy$  satisfies  $|z| \leq 1$ .

*Please, turn over!*

6. Consider the series

$$s(x) = \sum_{j=1}^{\infty} \frac{x^j + (1-x)^j}{j^2}.$$

- (a) Prove that  $s(x)$  is continuous on the closed interval  $[0, 1]$  and continuously differentiable on the open interval  $(0, 1)$ .
- (b) Find the derivative  $s'(x)$  when  $0 < x < 1$ .
- (c) Evaluate  $s(x)$  for  $0 < x < 1$ . (You may use that  $\sum_1^{\infty} \frac{1}{j^2} = \frac{\pi^2}{6}$ .)